

The Order-One Normalisations are Determined

$c'_S = \frac{1}{4}$, the cycle 2π , and the radiation constant (ledger D6)

Evaluation note — FRC 21-gravity

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Abstract

The body leaves three order-one numbers “open”: the entropy coefficient c'_S , the 2π of the acceleration floor, and the radiation constant. None is free. The coefficient c'_S is fixed to the Bekenstein–Hawking value $\frac{1}{4}$ by the de Sitter closure — the same area law, applied to the cosmological horizon, must reproduce the de Sitter entropy that defines the cardinality. The 2π of $a_0 = cH_0/2\pi$ is the angular period of one full phase turn, one cycle per Hubble time; the same 2π sets the Hawking temperature. The radiation constant is the spin-two quadrupole factor fixed by the radiative sector (D5). The recurring geometric constant throughout is the solid angle 4π . With these pinned, the construction is parameter-free in the literal sense.

1 $c'_S = \frac{1}{4}$ from the de Sitter closure

The horizon entropy is $S = c'_S A/\ell_P^2$ with A the area of the bounding surface. Two applications of the *one* law fix c'_S .

Proposition 1 (Operational horizon). *For the operationally black object the surface sits at $r_f = r_s/\ln \Omega$, area $A_f = 4\pi r_f^2$, so $S_{\text{op}} = c'_S 4\pi r_f^2/\ell_P^2$. The Bekenstein–Hawking entropy of a black hole of the same mass is $S_{\text{BH}} = \pi r_s^2/\ell_P^2$, so $S_{\text{op}}/S_{\text{BH}} = 4c'_S(r_f/r_s)^2$. The body’s stated ratio $S_{\text{op}}/S_{\text{BH}} = (r_f/r_s)^2 = \ln^{-2} \Omega$ therefore forces $4c'_S = 1$, i.e. $c'_S = \frac{1}{4}$.*

Proposition 2 (Cosmological horizon). *Applied to the Hubble horizon of radius $\sqrt{\Omega} \ell_P$, the area is $A = 4\pi \Omega \ell_P^2$, so the same law gives $S_{\text{cosmo}} = c'_S 4\pi \Omega$. The de Sitter entropy that fixes the cardinality is $S_{\text{dS}} = A/4\ell_P^2 = \pi \Omega$. The closure $S_{\text{cosmo}} = S_{\text{dS}}$ again gives $c'_S = \frac{1}{4}$.*

Two independent closures return the same value, the Bekenstein–Hawking–Gibbons–Hawking $\frac{1}{4}$. It is not derived from a microstate count — that is the standard hard problem of horizon thermodynamics — but it is *determined*, not fitted: the substrate’s own self-consistency (its information content is the de Sitter entropy) pins it, exactly as the Immirzi parameter is pinned in loop quantum gravity, but here by the cardinality rather than by hand.

2 The recurring constant is the solid angle 4π

The order-one factors of the construction are not independent. The same solid angle 4π appears in Newton’s constant $G = 1/4\pi\kappa$, the bare electromagnetic coupling $\alpha_{\text{bare}} = 1/4\pi$, the lattice Green’s function $\sim 1/4\pi r$, and the horizon area $4\pi r^2$; with $c'_S = \frac{1}{4}$ the entropy is $S = \pi r^2/\ell_P^2$. One geometric constant, 4π , runs through the gravitational, the electromagnetic, and the thermodynamic sectors.

3 The 2π of the floor is the cycle period

The floor $a_0 = cH_0/2\pi$ is the gradient that accumulates one phase cycle per Hubble time (Proposition `prop:afloor`). One phase cycle is a full turn, 2π radians: the 2π is the angular period of the phase circle, not a free coefficient. The scale cH_0 is the drive rate at the horizon; the 2π is the geometry of one revolution. The same 2π sets the Hawking temperature $k_B T \sim \hbar \kappa_{\text{sg}}/2\pi c$ (the Euclidean period of one thermal turn). Numerically $a_0 = 1.08 \times 10^{-10} \text{ m s}^{-2}$ ($H_0 = 70$) against the fitted 1.20×10^{-10} , within ten percent, with no parameter [approx].

4 The radiation constant is the spin-two quadrupole factor

The gravitational-wave luminosity $L = (G/5c^5)\langle \ddot{Q}_{ij}^2 \rangle$ has its coefficient fixed by the spin-two angular structure of the radiative sector (D5, the same linear wave theory as general relativity): the $\frac{1}{5}$ is the integral of the quadrupole pattern over the sphere, determined by the two transverse-traceless polarisations, not free.

Status. Ledger D6 moves from O to T | B: the order-one normalisations are determined — $c'_S = \frac{1}{4}$ by the de Sitter closure, the 2π as the cycle period, the radiation constant by the spin-two structure — and the recurring 4π is the solid angle. “Parameter-free” is then literal. Checks in `validation/validate_orderone.py`.